

Submission STAT1-FINAL-SUB08

Question STAT1-FINAL-Q01

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visible work:

$$\bar{x} = 26/5 = 5.20 \dots ?$$

last line is hard to read; maybe mean=5.20; sample variance=2.20

Question STAT1-FINAL-Q02

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visible work:

$$P(D_1 \cap D_2) = (4/15)((4-1)/(15-1)) \dots ?$$

last line is hard to read; maybe 0.0571

Question STAT1-FINAL-Q03

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visible work:

$$SE = \sqrt{(p(1-p)/n)} = \sqrt{(0.45(1-0.45)/40)} = 0.0787 \dots ?$$

last line is hard to read; maybe (0.296, 0.604); repeated-sampling interpretation

Question STAT1-FINAL-Q04

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visible work:

Increase sample size n to reduce standard error.... ?

last line is hard to read; maybe increase n; lower confidence level or variability; note the coverage trade-off

Question STAT1-FINAL-Q05

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visible work:

$0.03 < 0.05$, so reject H_0 under the stated rule.... ?

last line is hard to read; maybe reject H_0 ; evidence against H_0 , not proof that H_0 is false

Question STAT1-FINAL-Q06

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visible work:

$$\bar{x} = 31/5 = 6.20 \dots ?$$

last line is hard to read; maybe mean=6.20; sample variance=1.70

Question STAT1-FINAL-Q07

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visible work:

$$P(D_1 \cap D_2) = (5/19)((5-1)/(19-1)) \dots ?$$

last line is hard to read; maybe 0.0585

Question STAT1-FINAL-Q08

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visible work:

$$SE = \sqrt{p(1-p)/n} = \sqrt{(0.35(1-0.35)/80)} = 0.0533... ?$$

last line is hard to read; maybe (0.245, 0.455); repeated-sampling interpretation

Question STAT1-FINAL-Q09

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visible work:

Increase sample size n to reduce standard error.... ?

last line is hard to read; maybe increase n ; lower confidence level or variability; note the coverage trade-off

Question STAT1-FINAL-Q10

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visible work:

$0.03 < 0.05$, so reject H_0 under the stated rule.... ?

last line is hard to read; maybe reject H_0 ; evidence against H_0 , not proof that H_0 is false